Quiz 1

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1.) Consider the following permutations in S_7:

$$\sigma = (13)(256)$$
$$\tau = (417)(235)$$

What is the cycle type of σ and τ?

Solution:

We know that both permutations belong in S_7, and so there are 7 elements to be permuted. We also know that the cycle type of a permutation is a list representing the lengths of each of the permutation's cycles, placed in descending order. Using this, we can:

(I) Find the cycle type of σ.

σ has two explicitly written cycles: (13) and (256) which are of lengths 2 and 3 respectively. Since there are only 5 elements expressed in the cycle notation of σ, that mean that the other 2 elements of S_7, namely 4 and 7, were omitted for being fixed points. Because fixed points evidently have cycle lengths of 1, this leads us to conclude that the cycle type of σ must be given by:

$$(3, 2, 1, 1)$$

(II) Find the cycle type of τ.

By similar logic, τ has two cycles of length 3 each given by the notation. Since that makes for 6 elements total, the unaccounted for element, 6, must be fixed. Hence, the cycle type of τ is:

$$(3, 3, 1)$$

2.) How many elements are there in $A_4 \times D_4$.

Solution:

$$\begin{aligned} |A_4 \times D_4| &= |A_4|\,|D_4| \\ &= \left(\frac{4!}{2}\right)(8) \\ &= (12)(8) \\ &= 96 \end{aligned}$$

3.) Consider the following two permutations:

$$\alpha = (12)(34)(56)$$
$$\beta = (123)(456)$$

Which, if possible to deduce, of α and β are even or odd?

Solution:

4.) Which of the following sets of permutations belong to the group D_5? Hint: Try drawing a pentagon with vertices labelled clockwise 1 through 5. You are looking for adjacency permutations of this pentagon.

(a) $\alpha = (12)(34)$.

Solution:

As given by the hint, for α to belong to D_5, after permuting the following pentagon:

Node B

Node A